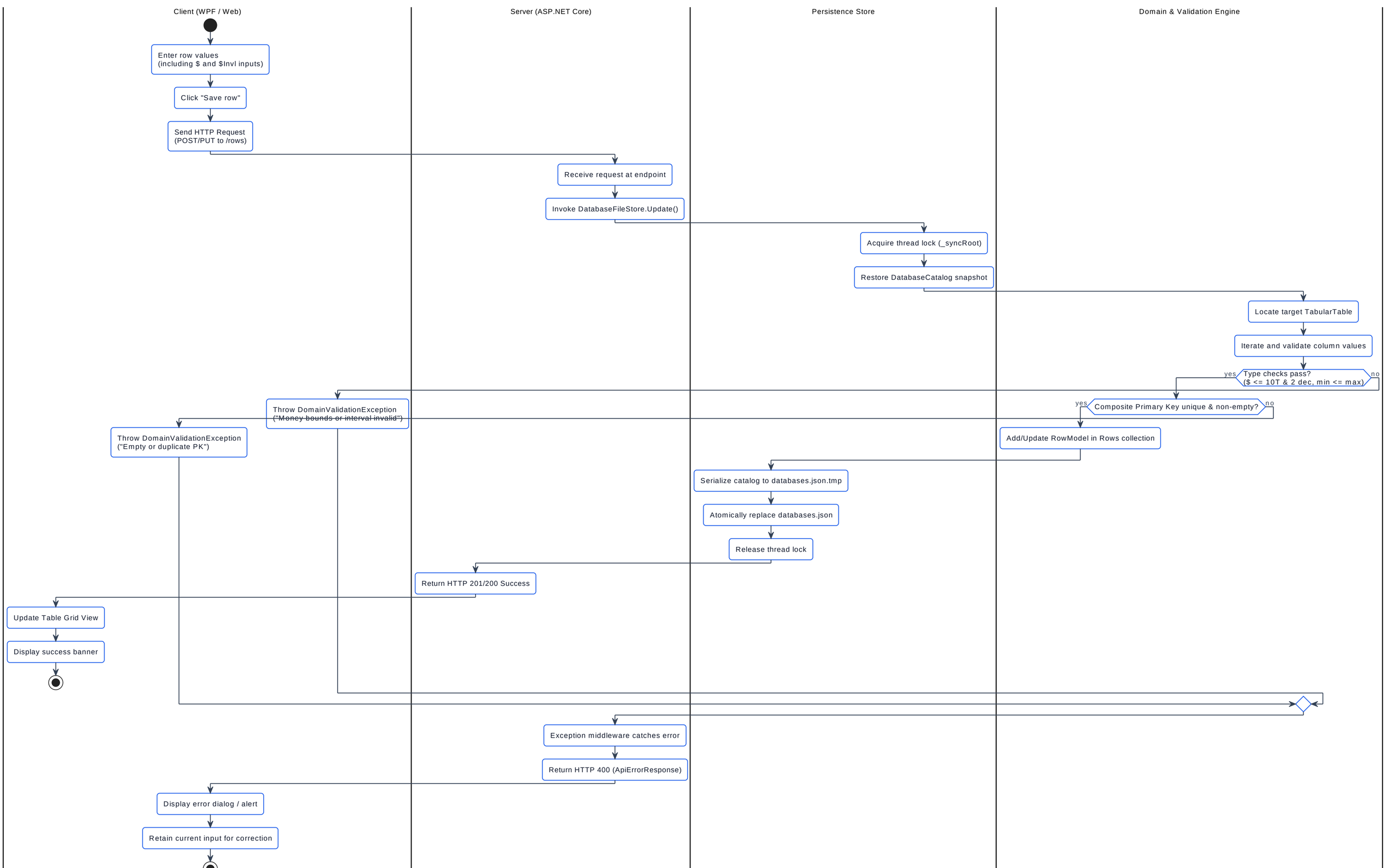
Client (WPF / Web)
Server (ASP.NET Core)
Persistence Store
Domain & Validation Engine
Enter row values
(including $ and $Invl inputs)
Click "Save row"
Send HTTP Request
(POST/PUT to /rows)
Receive request at endpoint
Invoke DatabaseFileStore.Update()
Acquire thread lock (_syncRoot)
Restore DatabaseCatalog snapshot
Locate target TabularTable
Iterate and validate column values
Type checks pass?
($ <= 10T & 2 dec, min <= max)
yes
no
Composite Primary Key unique & non-empty?
yes
no
Add/Update RowModel in Rows collection
Throw DomainValidationException
("Money bounds or interval invalid")
Throw DomainValidationException
("Empty or duplicate PK")
Serialize catalog to databases.json.tmp
Atomically replace databases.json
Release thread lock
Return HTTP 201/200 Success
Update Table Grid View
Display success banner
Exception middleware catches error
Return HTTP 400 (ApiErrorResponse)
Display error dialog / alert
Retain current input for correction